Pragmastat: Pragmatic Statistical Toolkit

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Abstract

This manual presents a unified statistical toolkit for reliable analysis of real-world data. The toolkit nearly matches the efficiency of traditional statistical estimators under normality, maintains reasonable robustness against outliers, enables simple software implementations without advanced statistical libraries, and provides clear explanations accessible to practitioners without deep statistical training. The toolkit consists of renamed, recombined, and refined versions of existing methods. Written for mathematicians, software developers, and LLMs.

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1 Introduction

The toolkit consists of statistical *procedures* (only estimators at the moment, but it will be extended with other methods, functions, and algorithms). These procedures process measurements and provide statistical insights or decisions. The practitioner's main task is choosing the right procedure for each problem. To build a pragmatic statistical toolkit, consider the following groups of desired properties.

Usability

- Be intuitive, with minimal risk of misuse
- Be mathematically elegant and simple, accessible to readers with average mathematical skills
- Be simple to implement in code (can be written from scratch in any language without special libraries or complex calculations)
- Have a small number of configuration parameters with reasonable defaults
- Follow the same spirit of usability as computer interfaces and everyday things; see [Nor13]

Reliability

- Provide reliable, precise, and useful results for real-world problems
- Provide consistent results for similar studies despite randomness
- Be robust to corrupted or extreme measurements
- Be location- and scale-invariant to ensure portability

Applicability

- Be applicable across diverse contexts
- Work well not only asymptotically but also with small sample sizes
- Work pretty well for the Normal distribution and reasonably well for other popular models: Uniform / Beta / Weibull / Student's t / Gumbel / Exponential / Cauchy / Pareto / Log-normal / Fréchet / etc.
- Work reliably across a wide range of assumptions:
  - Continuous / Discrete / Continuous-discrete mixtures
  - Uniform / Unimodal / Bimodal / Multimodal
  - Light-tailed / Exponential / Heavy-tailed
  - Symmetric / Right-skewed / Left-skewed
  - Independent / Dependent (correlated measurements)
  - Identically distributed / Heterogeneously distributed (weighted data)
- Tolerate slight deviations from popular assumptions
- Properly handle corner cases

This manual presents a toolkit of rule-of-thumb statistical procedures that naturally complement each other.

2 Preliminaries

This toolkit analyzes random variables X and Y. Each random variable produces random real numbers from its underlying distribution. Consider a set of n random values $\mathbf{X}$ and a set of m random values $\mathbf{Y}$:

$$\mathbf{X} = (X_1, X_2, \ldots, X_n)$$

$$\mathbf{Y} = (Y_1, Y_2, \ldots, Y_m)$$

These variables produce *measurements* (the actual observed values). For random values $\mathbf{X}$, the measurements are $x_1, x_2, \ldots, x_n$. Collections of measurements form *samples*:

$$\mathbf{x} = (x_1, x_2, \ldots, x_n)$$

$$\mathbf{y} = (y_1, y_2, \ldots, y_m)$$

The objective is to learn about the original random variables X and Y from these samples.

The *median* represents a fundamental property of any distribution. This value divides the distribution into two equal parts, creating exactly 50% probability of observing a measurement below the median and 50% above it. The *true* median value $\text{Median}(X)$ remains unknown, but the measurements provide an estimate.

To calculate the median, arrange the sample elements into an ordered sequence $x_{(1)}, \ldots, x_{(n)}$. This *sorted sample*[1] places $x_{(1)}$ as the smallest value and $x_{(n)}$ as the largest. The median estimate uses the middle value of this sorted sample. For even sample sizes, the median equals the average of the two middle values:

$$\text{Median}(\mathbf{x}) = \begin{cases} x_{((n+1)/2)} & \text{if } n \text{ is odd} \\ \frac{x_{(n/2)} + x_{(n/2+1)}}{2} & \text{if } n \text{ is even} \end{cases}$$

This estimation $\text{Median}(\mathbf{x})$ approximates the true distribution value $\text{Median}(X)$.

The median represents one example of a *quantile*. For probability $p \in [0, 1]$, the pth quantile $\text{Quantile}(X, p)$ divides the distribution so that the probability of obtaining a measurement below the quantile equals exactly p:

$$\Pr[X \leq \text{Quantile}(X, p)] = p$$

Linear interpolation between two sample elements produces quantile estimates[2]:

$$\begin{gathered} \text{Quantile}(\mathbf{x}, p) = x_{(\lfloor h \rfloor)} + (h - \lfloor h \rfloor) \cdot (x_{(\lceil h \rceil)} - x_{(\lfloor h \rfloor)}) \\ h = (n-1)p + 1 \end{gathered}$$

The median equals $\text{Quantile}(\mathbf{x}, 0.5)$ since it divides the distribution into equal halves. Median estimation produces higher precision than other quantile values. Estimation precision decreases as p approaches 0 or 1. Larger sample sizes improve quantile estimation precision for any value of p.

These quantile estimators form the building blocks for more complex distributional properties.

3 Estimators

In this section, we consider one-sample and two-sample estimators. These estimators can be used directly to obtain basic insights from the samples, or as a building block in composite procedures. Estimator evaluation focuses on these key properties:

- *Gaussian efficiency*: performance quality under normal distribution conditions. High efficiency produces accurate estimates with fewer measurements under normality.

[1] Also known as *order statistics*

[2] Definition uses the Hyndman-Fan Type 7 quantile estimator, see [HF96].

- *Breakdown point*: which portion of the sample can be replaced by corrupted measurements without impacting the estimation.
- *Invariance properties* (allow higher portability)
  - Shift-invariance: result is the same if all measurements are increased by the same constant
  - Scale-invariance: result is the same if all measurements are multiplied by the same constant
- *Domain*: Supported set of measurement values

3.1 Center

$$\mathrm{Center}(\mathbf{x}) = \underset{1 \le i \le j \le n}{\mathrm{Median}} \left(\frac{x_i + x_j}{2} \right)$$

- Measures central tendency (the average value)
- Provides initial insight into the magnitude of sample values
- When RelSpread is small, the entire sample can be approximated by the Center value
- Equals the *Hodges-Lehmann estimator* ([HL63], [Sen63]), renamed to "Center" for clarity
- Called *pseudomedian* in some texts because it is consistent with the median for symmetric distributions

Comparison

- Compared to the *mean*:
  - Robustness: tolerates almost one-third of outliers unlike the mean that can be corrupted by a single outlier
  - Efficiency: requires 4..10% more measurements for the same precision
- Compared to the *median*:
  - Robustness: lower than the median, which tolerates half of the samples corrupted (although, such extreme robustness rarely provides practical advantage)
  - Efficiency: requires 1.5 times fewer measurements for the same precision

Properties

- Domain: any real numbers
- Unit: the same as measurements
- Asymptotic Gaussian efficiency: $\approx 96\%$
- Finite-sample Gaussian efficiency $> 91\%$
- Asymptotic breakdown point: $\approx 29\%$ (matches Spread)
- Location-dependent, scale-dependent

$$\mathrm{Center}(\mathbf{x} + k) = \mathrm{Center}(\mathbf{x}) + k$$

$$\mathrm{Center}(k \cdot \mathbf{x}) = k \cdot \mathrm{Center}(\mathbf{x})$$

3.2 Spread

$$\text{Spread}(\mathbf{x}) = \underset{1 \le i < j \le n}{\text{Median}} |x_i - x_j|$$

- Measures dispersion (variability, scatter)
- Provides insight into the median absolute difference between two random measurements
- Interpretation: half of $|x_i - x_j|$ values are smaller than $\text{Spread}(\mathbf{x})$, the other half are larger
- Corner case: for $n = 1$, use $\text{Spread}(\mathbf{x}) = 0$
- Equals the *Shamos scale estimator* ([Sha76]), renamed to "Spread" for clarity

Comparison

- Compared to the *standard deviation*:
  - Robustness: the spread tolerates almost one-third of outliers unlike the standard deviation that can be corrupted by a single outlier
  - Efficiency: good (asymptotic relative efficiency of the spread to the standard deviation is of $\approx 86\%$)
  - Usability: More intuitive without requiring knowledge of normal distributions
  - Consistency: no consistency under normality without multiplication by a constant
- Compared to the *median absolute deviation*:
  - Robustness: median tolerates half of the samples corrupted, which is more than one-third for the spread, but it does not provide practical advantage
  - Efficiency: requires ≈ 2.35 times fewer observations for the same precision

Spread can be converted to an unbiased consistent estimator for another dispersion measure by multiplication by a constant

Properties

- Domain: any real numbers
- Unit: the same as measurements
- Location-invariant, scale-dependent
- Asymptotic Gaussian efficiency: $\approx 86\%$
- Asymptotic breakdown point: $\approx 29\%$ (matches Center in robustness)
- Asymptotic expected value for the standard normal distribution: ≈ 0.954
- Coverage for the normal distribution:
  - $[\text{Center}(\mathbf{x}) \pm 1 \cdot \text{Spread}(\mathbf{x})]$ covers $\approx 65.98518\%$ of the normal distribution
  - $[\text{Center}(\mathbf{x}) \pm 2 \cdot \text{Spread}(\mathbf{x})]$ covers $\approx 94.35758\%$ of the normal distribution
  - $[\text{Center}(\mathbf{x}) \pm 3 \cdot \text{Spread}(\mathbf{x})]$ covers $\approx 99.57851\%$ of the normal distribution
  - $[\text{Center}(\mathbf{x}) \pm 4 \cdot \text{Spread}(\mathbf{x})]$ covers $\approx 99.98641\%$ of the normal distribution
  - $[\text{Center}(\mathbf{x}) \pm 5 \cdot \text{Spread}(\mathbf{x})]$ covers $\approx 99.99982\%$ of the normal distribution

$$\text{Spread}(\mathbf{x} + k) = \text{Spread}(\mathbf{x})$$

$$\text{Spread}(k \cdot \mathbf{x}) = |k| \cdot \text{Spread}(\mathbf{x})$$

$$\text{Spread}(\mathbf{x}) \ge 0$$

3.3 RelSpread

$$\text{RelSpread}(\mathbf{x}) = \frac{\text{Spread}(\mathbf{x})}{|\text{Center}(\mathbf{x})|}$$

- Measures the relative dispersion of a sample to $\text{Center}(\mathbf{x})$
- Interpretation examples:
  - $\text{RelSpread}(\mathbf{x}) = 0.01$: data clusters tightly around $\text{Center}(\mathbf{x})$ with minimal variation
  - $\text{RelSpread}(\mathbf{x}) = 0.1$: moderate variation, typical values range from 90% to 110% of center
  - $\text{RelSpread}(\mathbf{x}) = 1.0$: high variation, values span from near zero to twice the center

Comparison

- Robust alternative to the *coefficient of variation*

Properties

- Mathematical Domain: $\text{Center}(\mathbf{x}) \neq 0$
- Pragmatic Domain: non-negative values allowing up to 29% zeros
- Unit: relative (dimensionless)
- Location-dependent, scale-invariant

$$\text{RelSpread}(k \cdot \mathbf{x}) = \text{RelSpread}(\mathbf{x})$$

$$\text{RelSpread}(\mathbf{x}) \geq 0$$

3.4 Shift

$$\text{Shift}(\mathbf{x}, \mathbf{y}) = \underset{1 \leq i \leq n,\ 1 \leq j \leq m}{\text{Median}} (x_i - y_j)$$

- Measures the median difference between elements of two samples
- Value interpretation: half of $x_i - y_j$ values are smaller than $\text{Shift}(\mathbf{x}, \mathbf{y})$, the other half are larger
- Sign interpretation: positive shifts indicate larger $\mathbf{x}$ values, negative shifts indicate larger $\mathbf{y}$ values
- For samples with small RelSpread, $\text{Shift}(\mathbf{x}, \mathbf{y})$ approximates pairwise differences $x_i - y_j$
- Equals the *Hodges-Lehmann estimator* for two samples ([HL63])

Properties

- Domain: any real numbers
- Unit: the same as measurements
- Location-invariant, scale-dependent

$$\text{Shift}(\mathbf{x}, \mathbf{x}) = 0$$

$$\text{Shift}(\mathbf{x} + k_x, \mathbf{y} + k_y) = \text{Shift}(\mathbf{x}, \mathbf{y}) + k_x - k_y$$

$$\text{Shift}(k \cdot \mathbf{x}, k \cdot \mathbf{y}) = k \cdot \text{Shift}(\mathbf{x}, \mathbf{y})$$

$$\text{Shift}(\mathbf{x}, \mathbf{y}) = -\,\text{Shift}(\mathbf{y}, \mathbf{x})$$

3.5 Ratio

$$\mathrm{Ratio}(\mathbf{x}, \mathbf{y}) = \underset{1 \le i \le n,\ 1 \le j \le m}{\mathrm{Median}} \left(\frac{x_i}{y_j} \right)$$

- Measures the median ratio between elements of two samples as a multiplicative factor
- Second sample y is always the baseline
- In general, $\mathrm{Ratio}(\mathbf{x}, \mathbf{y}) \neq 1/\,\mathrm{Ratio}(\mathbf{y}, \mathbf{x})$ (example: $x = (1, 100)$, $y = (1, 10)$)
- Results convert to percentage differences as $(\mathrm{Ratio} - 1) \times 100\%$
- Interpretation example: Ratio = 2.0 means that for 50% of pairs (x_i, y_j), x_i is at least twice as large as y_j

Properties

- Domain: $y_j > 0$
- Unit: relative (dimensionless)
- Location-dependent, scale-invariant

$$\mathrm{Ratio}(k_x \cdot \mathbf{x}, k_y \cdot \mathbf{y}) = \frac{k_x}{k_y} \cdot \mathrm{Ratio}(\mathbf{x}, \mathbf{y})$$

3.6 AvgSpread

$$\mathrm{AvgSpread}(\mathbf{x}, \mathbf{y}) = \frac{n\,\mathrm{Spread}(\mathbf{x}) + m\,\mathrm{Spread}(\mathbf{y})}{n + m}$$

- Measures average dispersion across two samples
- Serves as a normalization factor for Disparity
- Performs best for distributions with close dispersion values

Comparison

- Robust alternative to the *pooled standard deviation*

Properties

- Domain: any real numbers
- Unit: the same as measurements
- Location-invariant, scale-invariant

$$\mathrm{AvgSpread}(\mathbf{x}, \mathbf{x}) = \mathrm{Spread}(\mathbf{x})$$

$$\mathrm{AvgSpread}(k_1 \cdot \mathbf{x}, k_2 \cdot \mathbf{x}) = \frac{|k_1| + |k_2|}{2} \cdot \mathrm{Spread}(\mathbf{x})$$

$$\mathrm{AvgSpread}(\mathbf{x}, \mathbf{y}) = \mathrm{AvgSpread}(\mathbf{y}, \mathbf{x})$$

$$\mathrm{AvgSpread}(k \cdot \mathbf{x}, k \cdot \mathbf{y}) = |k| \cdot \mathrm{AvgSpread}(\mathbf{x}, \mathbf{y})$$

3.7 Disparity

$$\text{Disparity}(\mathbf{x}, \mathbf{y}) = \frac{\text{Shift}(\mathbf{x}, \mathbf{y})}{\text{AvgSpread}(\mathbf{x}, \mathbf{y})}$$

- Measures a normalized difference between $\mathbf{x}$ and $\mathbf{y}$ expressed in standardized spread units
- Provides insight into the difference between elements of two samples, expressed in standardized spread units
- Expresses the *effect size*, renamed to "Disparity" for clarity

Comparison

- Compared to *Cohen's d*:
  - Gaussian efficiency: good
  - Robustness: tolerates a decent portion of outliers unlike Cohen's d which can be corrupted by a single outlier
- Compared to Shift: scale-invariant
- Compared to Ratio: shift-invariant

Properties

- Domain: $\text{AvgSpread}(\mathbf{x}, \mathbf{y}) > 0$ (in each sample, the portion of tied values does not exceed 50%)
- Unit: abstract spread unit
- Location-invariant, scale-invariant

$$\text{Disparity}(\mathbf{x} + k, \mathbf{y} + k) = \text{Disparity}(\mathbf{x}, \mathbf{y})$$

$$\text{Disparity}(k \cdot \mathbf{x}, k \cdot \mathbf{y}) = \text{sign}(k) \cdot \text{Disparity}(\mathbf{x}, \mathbf{y})$$

$$\text{Disparity}(\mathbf{x}, \mathbf{y}) = -\,\text{Disparity}(\mathbf{y}, \mathbf{x})$$

4 Studies

This section analyzes the estimators' properties using mathematical proofs and Monte Carlo simulations. Most proofs are adopted from various textbooks and research papers, but only essential references are provided.

Unlike the main part of the manual, the studies require knowledge of classic statistical methods. Well-known facts and commonly accepted notation are used without special introduction. The studies provide detailed analysis of estimator properties for practitioners interested in rigorous proofs and numerical simulation results.

4.1 Asymptotic Gaussian Expected Value of the Spread

This study establishes that Spread has expected value $\sqrt{2}\,\Phi^{-1}(0.75) \approx 0.954$ under standard normal data as sample size increases.

The key insight: pairwise absolute differences $|X_i - X_j|$ from normal data converge to a known distribution. Since Spread takes the median of these differences, its asymptotic expectation equals the population median of $|X_1 - X_2|$ where $X_1, X_2 \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$.

Consider $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$. For any fixed $i \neq j$, the difference $X_i - X_j$ has mean 0 and variance

$$\mathbb{V}[X_i - X_j] = \mathbb{V}[X_i] + \mathbb{V}[X_j] = 1 + 1 = 2$$

because X_i and X_j are independent. Therefore

$$X_i - X_j \sim \mathcal{N}(0,2)$$

Define $W = (X_i - X_j)/\sqrt{2} \sim \mathcal{N}(0,1)$ and $D = |X_i - X_j| = \sqrt{2}\,|W|$. The cumulative distribution function of D for $t \geq 0$ is

$$\Pr[D \leq t] = \Pr\big[|W| \leq t/\sqrt{2}\big] = 2\Phi\big(t/\sqrt{2}\big) - 1$$

where Φ denotes the standard normal cumulative distribution function. The population median m of D satisfies

$$2\Phi\big(m/\sqrt{2}\big) - 1 = \tfrac{1}{2}$$

Solving for m yields

$$m = \sqrt{2}\,\Phi^{-1}(0.75)$$

The multiset of gaps $\{|X_i - X_j| : i < j\}$ forms a bounded-kernel U-statistic of degree 2. U-statistic consistency results imply that its empirical distribution converges almost surely to the law of D. The sample median of pairwise differences thus converges in probability to m. Convergence in probability plus uniform integrability yields convergence of expectations:

$$\lim_{n\to\infty} \mathbb{E}\big[\text{Spread}(X_1, \ldots, X_n)\big] = \sqrt{2}\,\Phi^{-1}(0.75)$$

Numerically, this equals

$$\lim_{n\to\infty} \mathbb{E}\big[\text{Spread}(X_1, \ldots, X_n)\big] \approx 0.953\,873$$

4.2 Asymptotic Gaussian Efficiency of the Center

This study shows that Center achieves $3/\pi \approx 95.5\%$ efficiency relative to the sample mean under normality. This high efficiency allows Center to handle outliers while maintaining near-optimal performance on normal data.

The analysis uses U-statistic theory to derive the asymptotic distribution of Center under Gaussian data. Let

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2), \qquad n \geq 2,\ \sigma > 0$$

Center($\mathbf{x}$) has translation invariance, so setting $\mu = 0$ preserves generality. The Walsh averages are

$$W_{ij} = \frac{X_i + X_j}{2}, \qquad 1 \leq i \leq j \leq n$$

Since $X_i + X_j \sim \mathcal{N}(0, 2\sigma^2)$, it follows that

$$W_{ij} \sim \mathcal{N}\big(0, \sigma^2/2\big), \qquad f_W(0) = \frac{1}{\sigma\sqrt{\pi}}$$

The estimator Center($\mathbf{x}$) equals the sample median of the $\binom{n+1}{2}$ Walsh averages. As a U-quantile of degree two, it satisfies

$$0 = \frac{2}{n(n+1)} \sum_{1 \leq i \leq j \leq n} \Big\{ \mathbf{1}\{W_{ij} \leq \text{Center}(\mathbf{x})\} - \tfrac{1}{2} \Big\}$$

U-quantile theory [Sen63] provides the linear expansion

$$\sqrt{n}\ \text{Center}(\mathbf{x}) = \frac{2}{f_W(0)} \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi(X_i) + o_{\Pr}(1)$$

where

$$\psi(x) = \Pr\Big\{ \tfrac{x + X_2}{2} \leq 0 \Big\} - \tfrac{1}{2} = \tfrac{1}{2} - \Phi\Big(\tfrac{x}{\sigma}\Big)$$

and Φ is the standard normal cumulative distribution function. Since $X_1/\sigma \sim \mathcal{N}(0, 1)$,

$$\mathbb{V}\big[\psi(X_1)\big] = \int_{-\infty}^{\infty} \Big(\tfrac{1}{2} - \Phi(u)\Big)^2 \varphi(u)\, du = \frac{1}{12}$$

with φ the standard normal probability density function. Substituting $\mathbb{V}[\psi(X_1)]$ and $f_W(0)$ yields

$$\mathbb{V}\big[\sqrt{n}\ \text{Center}(\mathbf{x})\big] = \frac{4 \cdot \frac{1}{12}}{\big(1/(\sigma\sqrt{\pi})\big)^2} = \frac{\pi\sigma^2}{3}$$

so

$$\sqrt{n}\ \text{Center}(\mathbf{x}) \xrightarrow{d} \mathcal{N}\Big(0, \tfrac{\pi\sigma^2}{3}\Big), \qquad n \to \infty$$

The sample mean has asymptotic variance σ^2/n, hence the *asymptotic Gaussian efficiency* of Center is

$$\text{eff}_{\mathcal{N},\infty}(\text{Center}) = \frac{\sigma^2/n}{\pi\sigma^2/(3n)} = \frac{3}{\pi} \approx 0.954\,930$$

Thus matching the mean's precision under normality requires only $1/0.955 \approx 1.05$ times as many observations, a negligible price for Center's 29% breakdown point and much stronger resistance to outliers.

4.3 Asymptotic Gaussian Efficiency of the Median

This study shows that the sample median achieves only $2/\pi \approx 63.7\%$ efficiency relative to the sample mean under normality.

The analysis applies classical asymptotic theory for sample quantiles to derive the limiting distribution of the median under Gaussian data. Consider

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2), \qquad n \geq 2,\ \sigma > 0$$

Since the sample median is translation invariant, set $\mu = 0$ without loss of generality. Write

$$M_n = \text{Median}(X_1, \ldots, X_n)$$

For any continuous distribution with density f positive at its median θ, classical theory ([SSH99], [Ser09]) gives

$$\sqrt{n}\,(M_n - \theta) \xrightarrow{d} \mathcal{N}\Big(0, \frac{1}{4f(\theta)^2}\Big)$$

In the normal case $\theta = 0$ and

$$f(0) = \frac{1}{\sigma\sqrt{2\pi}}$$

so the asymptotic variance becomes

$$\frac{1}{4f(0)^2} = \frac{1}{4}\left(\sigma\sqrt{2\pi}\right)^2 = \frac{\pi\sigma^2}{2}$$

Hence

$$\sqrt{n}\,M_n \xrightarrow{d} \mathcal{N}\Big(0, \frac{\pi\sigma^2}{2}\Big), \qquad n \to \infty$$

The sample mean has asymptotic variance σ^2/n, so the *asymptotic Gaussian efficiency* of the median is

$$\text{eff}_{\mathcal{N},\infty}(\text{Median}) = \frac{\sigma^2/n}{\pi\sigma^2/(2n)} = \frac{2}{\pi} \approx 0.637$$

Thus achieving the same precision as the mean under normality requires $1/0.637 \approx 1.57$ times as many observations when using the median. This large efficiency loss shows why the analysis prefers the Hodges–Lehmann Center estimator, which attains about 95.5% efficiency, whenever data are roughly Gaussian but may include outliers.

4.4 Asymptotic Gaussian Efficiency of the Spread

This study shows that Spread achieves approximately 86% efficiency relative to the sample standard deviation under normality.

The analysis uses U-statistic theory to derive the asymptotic distribution of Spread as a scale estimator under Gaussian data. Consider

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2), \qquad n \geq 2,\ \sigma > 0$$

Since $\text{Spread}(\mathbf{x}) = \text{Median}\,|X_i - X_j|$ is translation invariant, set $\mu = 0$ without loss of generality and write

$$R_n = \text{Spread}(\mathbf{x})$$

$$m_0 = \sqrt{2}\,\Phi^{-1}(0.75)\,\sigma \approx 0.954\,\sigma$$

Here m_0 denotes the population median of $|X_i - X_j|$ when $X_i \sim \mathcal{N}(0, \sigma^2)$. Letting $D = |X_1 - X_2|$, its density is

$$f_D(t) = \frac{1}{\sigma\sqrt{\pi}}\,\exp\Big(-\frac{t^2}{4\sigma^2}\Big), \qquad t \geq 0$$

with $f_D(m_0) = \sigma^{-1}\pi^{-1/2}\exp(-\Phi^{-1}(0.75)^2/2)$. Treating R_n as a degree-two U-quantile and applying asymptotic theory [Sen63] yields

$$\sqrt{n}\,\big(R_n - m_0\big) \xrightarrow{d} \mathcal{N}\Big(0, \frac{4\,\mathbb{V}[\psi(X_1)]}{f_D(m_0)^2}\Big)$$

where

$$\begin{aligned}\psi(x) &= \Pr\big\{|x - X_2| \leq m_0\big\} - \tfrac{1}{2} \\ &= \Phi\Big(\tfrac{x+m_0}{\sigma}\Big) - \Phi\Big(\tfrac{x-m_0}{\sigma}\Big) - \tfrac{1}{2}\end{aligned}$$

Numerical evaluation of the integral

$$\mathbb{V}[\psi(X_1)] = \int_{-\infty}^{\infty} \psi(x)^2\,\frac{e^{-x^2/(2\sigma^2)}}{\sigma\sqrt{2\pi}}\,dx$$

yields $\mathbb{V}[\psi(X_1)] \approx 0.0266$. Substituting this value and $f_D(m_0)$ into the variance formula gives

$$\mathbb{V}\big[\sqrt{n}\,R_n\big] \approx 0.527\,\sigma^2$$

Because R_n is *not* a consistent estimator of σ, comparisons with the sample standard deviation $\text{StdDev}(\mathbf{x})$ use the rescaled statistic

$$\widehat{\sigma}_{\text{Spread}} = \frac{R_n}{m_0}$$

which *is* consistent. Dividing the variance above by the constant m_0^2 gives

$$\mathbb{V}\big[\sqrt{n}\,\widehat{\sigma}_{\text{Spread}}\big] \approx 0.579\,\sigma^2$$

The optimal Gaussian scale estimator $\text{StdDev}(\mathbf{x})$ has asymptotic variance $\sigma^2/(2n)$, so the *asymptotic Gaussian efficiency* of the (scaled) Spread is

$$\text{eff}_{\mathcal{N},\infty}(\text{Spread}) = \frac{\sigma^2/(2n)}{0.579\,\sigma^2/n} \approx 0.864$$

One needs roughly $1/0.864 \approx 1.16$ times as many observations to match the precision of the sample standard deviation when the data are exactly normal. In exchange, Spread inherits a 29% breakdown point from its U-quantile construction, so moderately extra data provide a substantial increase in robustness.

4.5 Asymptotic Gaussian Efficiency of the Median Absolute Deviation

This study shows that the median absolute deviation (MAD) achieves only about 37% efficiency relative to the sample standard deviation under normality.

The analysis derives the asymptotic distribution of the MAD under Gaussian data using classical theory for sample medians. Consider

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2), \qquad n \geq 2,\ \sigma > 0$$

Since both the sample median and the MAD are translation invariant, set $\mu = 0$ without loss of generality and write

$$\text{MAD}_n = \text{Median}\Big(|X_i - \text{Median}(\mathbf{x})| : i = 1, \ldots, n\Big)$$

For a standard normal distribution, the population MAD equals

$$m_0 = \Phi^{-1}(0.75)\,\sigma = c_0\,\sigma$$

$$c_0 = \Phi^{-1}(0.75) \approx 0.674$$

Dividing the empirical MAD by this constant gives a consistent scale estimator:

$$\widehat{\sigma}_{\text{MAD}} = \frac{\text{MAD}_n}{c_0}$$

To find the large-sample variance, observe that $Y_i = |X_i|$ has density

$$g(y) = \frac{2}{\sigma\sqrt{2\pi}} \exp\Big(-\frac{y^2}{2\sigma^2}\Big), \qquad y \geq 0$$

whose median is m_0. A classical result for sample medians of independent draws with continuous positive density at the median ([SSH99], [Ser09]) states

$$\sqrt{n}\big(\text{Median}(Y_1, \ldots, Y_n) - m_0\big) \xrightarrow{d} \mathcal{N}\Big(0, \tfrac{1}{4g(m_0)^2}\Big)$$

Since

$$g(m_0) = \frac{2}{\sigma\sqrt{2\pi}} \exp\Big(-\frac{c_0^2}{2}\Big)$$

the asymptotic variance of $\sqrt{n}\ \mathrm{MAD}_n$ is

$$\mathbb{V}\big[\sqrt{n}\ \mathrm{MAD}_n\big] = \frac{1}{4g(m_0)^2} = \frac{\pi\sigma^2 \exp(c_0^2)}{8}$$

Scaling by $1/c_0$ gives the variance of the consistent estimator:

$$\mathbb{V}\big[\sqrt{n}\,\widehat{\sigma}_{\mathrm{MAD}}\big] = \frac{\pi\sigma^2 \exp(c_0^2)}{8\,c_0^2}$$

The optimal Gaussian scale estimator is the sample standard deviation. It has asymptotic variance of $\sigma^2/(2n)$. The *asymptotic Gaussian efficiency* of the (scaled) MAD is therefore

$$\mathrm{eff}_{\mathcal{N},\infty}(\mathrm{MAD}) = \frac{\sigma^2/(2n)}{\pi\sigma^2 \exp(c_0^2)/(8c_0^2 n)} = \frac{4c_0^2}{\pi\ \exp(c_0^2)}$$

This gives the result:

$$\mathrm{eff}_{\mathcal{N},\infty}(\mathrm{MAD}) \approx 0.367\,523$$

Achieving the same precision as the sample standard deviation under normality requires $1/0.368 \approx 2.7$ times as many observations when using the MAD.

4.6 Finite-Sample Efficiency of Central Tendency Estimators

This study presents finite-sample efficiency values for Center and shows how it performs better than Median across small and medium sample sizes.

The previous studies established asymptotic efficiency values (the limiting behavior as sample size approaches infinity). For the Gaussian distribution, these asymptotic values are:

- Mean: 100% (the most efficient estimator under normality)
- Center: $3/\pi \approx 95.5\%$
- Median: $2/\pi \approx 63.7\%$

Asymptotic theory provides excellent approximations for large samples but can be inaccurate for small n. Finite-sample efficiency captures the actual precision when working with limited data.

Efficiency and Sample Size Requirements

Efficiency quantifies the variance ratio between estimators. For two estimators T_1 and T_2 applied to the same distribution:

$$\mathrm{eff}(T_1 \text{ relative to } T_2) = \frac{\mathbb{V}[T_2]}{\mathbb{V}[T_1]}$$

This ratio directly translates to sample size requirements. An estimator with 80% efficiency needs $100/80 = 1.25$ times as many observations to achieve the same precision as the reference estimator. The Median with its 63.7% asymptotic efficiency requires roughly 1.57 times more data than the Mean under normality.

Monte Carlo Estimation of Finite-Sample Efficiency

Numerical simulation provides exact efficiency values for any sample size. The procedure follows these steps:

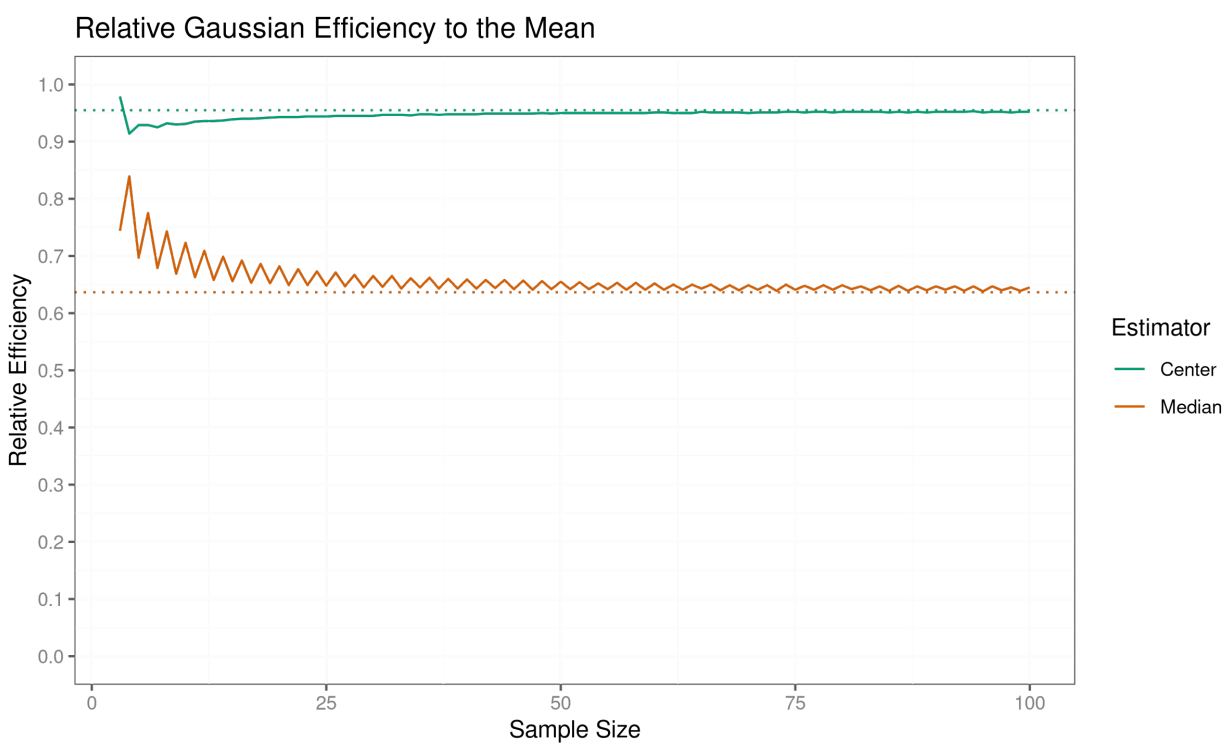
1. **Generate samples**: Draw m independent samples of size n from the standard normal distribution
2. **Calculate estimators**: Compute both estimators for each sample
3. **Measure dispersion**: Calculate the sample variance of the m estimator values
4. **Compute efficiency**: Take the variance ratio

The simulation must balance computational cost against precision. Larger m reduces Monte Carlo error but increases computation time. For efficiency estimation, $m = 10^6$ iterations achieve sufficient precision.

Finite-Sample Results

The simulation reveals how efficiency evolves from small to moderate sample sizes.

The figure below shows the Gaussian efficiency curves for $n \in \{3, \ldots, 100\}$ based on $m = 10^6$ Monte Carlo iterations (dotted lines show asymptotic values):



Key Observations

1. **Small sample behavior**: For $n = 3$, Center achieves 97.9% efficiency while Median drops to 74.3%. Even at extremely small samples, Center maintains above 91% efficiency throughout.

2. **Convergence patterns**: The Center estimator reaches 94% efficiency by $n = 15$ and stabilizes above 95% for $n \geq 50$. The Median oscillates between 63.5% and 68% across all sample sizes, never approaching the efficiency of Center.

3. **Practical advantage**: For typical applications ($n = 20$ to 50), Center maintains $94 - 95\%$ efficiency. This represents only a $5 - 6\%$ penalty compared to Mean, while Median suffers a $35 - 37\%$ penalty.

The toolkit's Center estimator performs much better than Median while sacrificing minimal efficiency compared to the non-robust Mean.

4.7 Finite-Sample Efficiency of Dispersion Estimators

This study presents finite-sample efficiency values for Spread and demonstrates its consistent superiority over the standard deviation StdDev and the median absolute deviation MAD across small and medium sample sizes.

The previous studies established asymptotic efficiency values for dispersion estimators under the Gaussian distribution. These asymptotic values are:

- StdDev: 100% (the most efficient estimator under normality)
- Spread: $\approx 86.4\%$
- MAD: $\approx 36.8\%$

The asymptotic values provide excellent approximations for large samples but may not accurately reflect performance with limited data. Understanding actual efficiency at practical sample sizes guides estimator selection from the toolkit.

Efficiency Measurement for Dispersion Estimators

Dispersion estimators face a variance-bias trade-off: different estimators have different expected values under the same distribution. To compare their variability fairly, the simulation normalizes each estimator by its expected value. This normalization eliminates bias differences and focuses purely on variance comparison.

The procedure follows these steps:

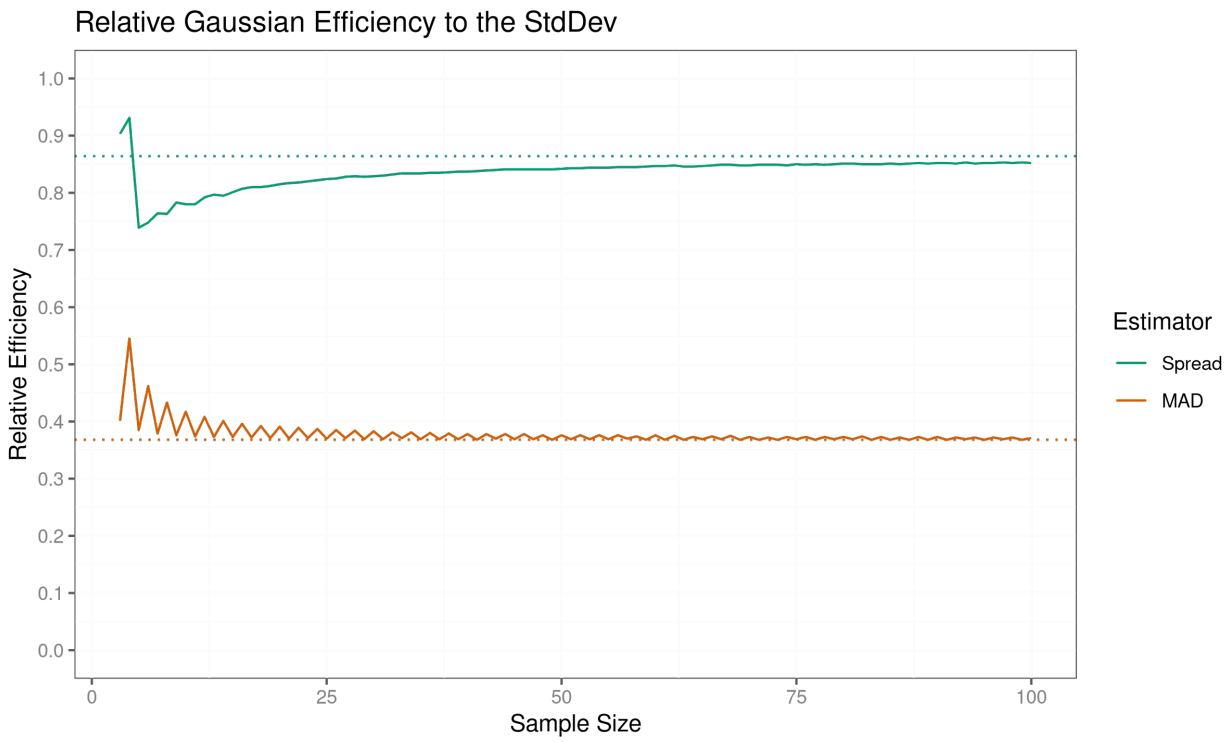
1. **Generate samples**: Draw m independent samples of size n from the standard normal distribution
2. **Calculate estimators**: Compute StdDev, Spread, and MAD for each sample
3. **Calculate normalized variance**: For each set of estimations, calculate the variance and divide by the mean value to align estimator biases
4. **Compute efficiency**: Take the variance ratio relative to StdDev

This approach ensures fair comparison by measuring how much each estimator varies around its own expected value. The simulation uses $m = 10^6$ iterations to achieve sufficient precision.

Results and Analysis

The simulation covers sample sizes $n \in \{3, \ldots, 100\}$ with efficiency computed relative to the standard deviation.

The below figure shows the Gaussian efficiency curves based on 10^6 Monte Carlo iterations (dotted lines show asymptotic values):



Key Observations

1. **Superior efficiency**: Spread consistently outperforms MAD by a factor of two across all sample sizes. At $n = 20$, Spread achieves 81.4% efficiency while MAD reaches only 39.0%.
2. **Practical implications**: For typical sample sizes ($n = 20$ to 50), Spread maintains $81-84\%$ efficiency. This $16-19\%$ penalty compared to StdDev represents a reasonable trade-off for robustness.
3. **Small sample behavior**: Spread shows a characteristic dip for $n = 5-8$ (dropping to 74%) before recovering. Even at its worst, Spread remains twice as efficient as MAD.

Spread from the toolkit provides a substantial efficiency advantage over MAD while maintaining reasonable performance compared to the non-robust standard deviation. This efficiency advantage, combined with its robustness properties, makes Spread the preferred choice for dispersion estimation in practical applications.

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